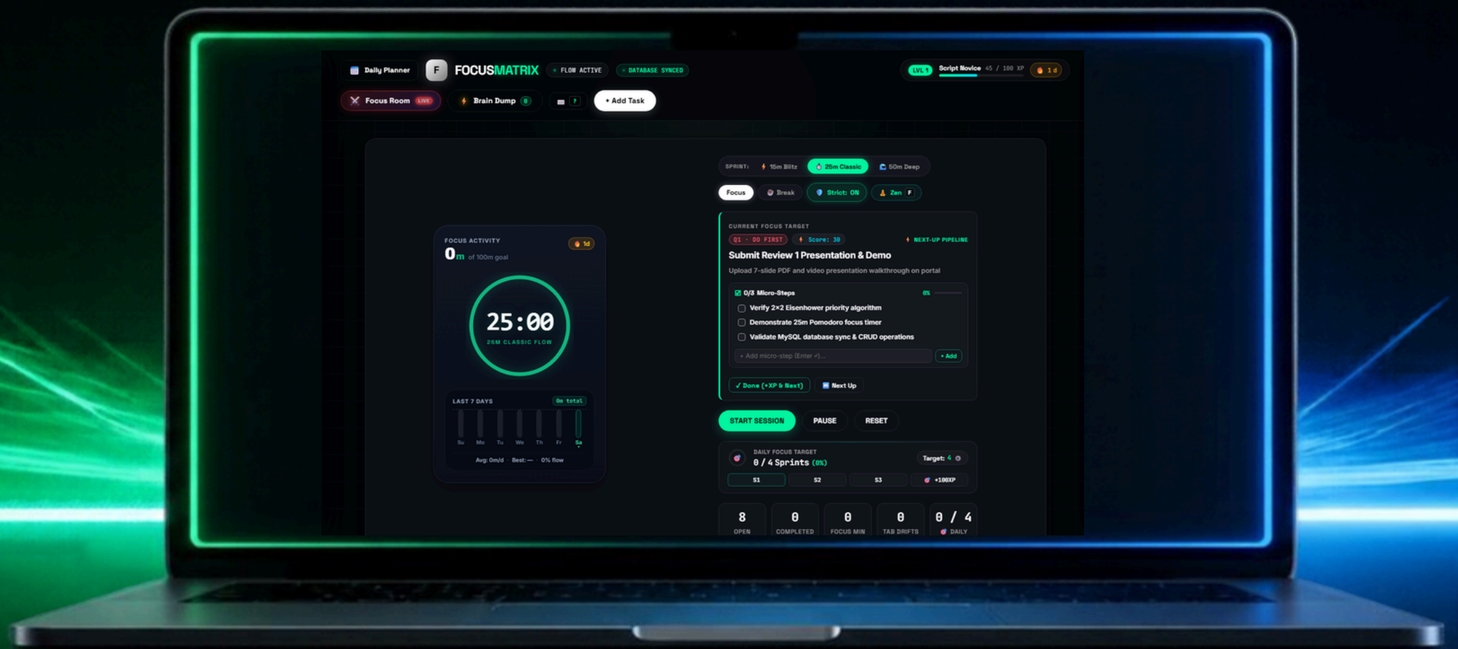
FocusMatrix

Stop Managing Lists. Start Crushing Work.



Team Byte Builder | Topic #63 | Galgotias University SCSE

Harsh Kumar · Aryan Singh · Anshu Kumar · Alakh Niranjan

Why Traditional To-Do Lists Are Broken

The Pain

Equal-Weight Trap:
5m chore = 20h project

Decision Fatigue:
15 mins wasted deciding what to start

Zero Focus Guard:
Lists don’t stop endless tab switching

FocusMatrix Fix

Algorithmic Triage:
Math prioritizes for you

Next-Up Queue:
One clear target at a time

Pomodoro Lockdown:
25-minute focused work cycles

The 2x2 Eisenhower Decision Engine

How the algorithm categorizes work automatically

Q1 — DO FIRST
Urgent & Critical
immediate deadlines

Q2 — SCHEDULE
High-Impact Deep Work
DSA, System Architecture

Q3 — BATCH
Quick Admin & Chores
Team WhatsApp syncs

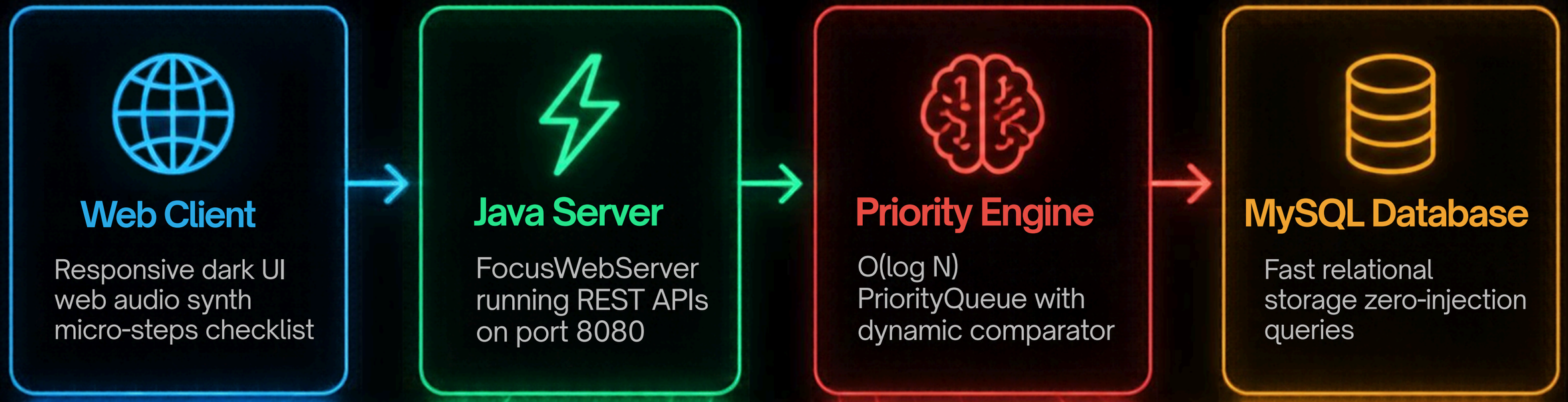
Q4 — ELIMINATE
Distractions & Clutter
Inbox cleaning

Priority Score = (Importance × 3) + [10 ÷ (Days + 1)] + (Urgency × 2)

Deadlines <24h
auto-escalate to Q1

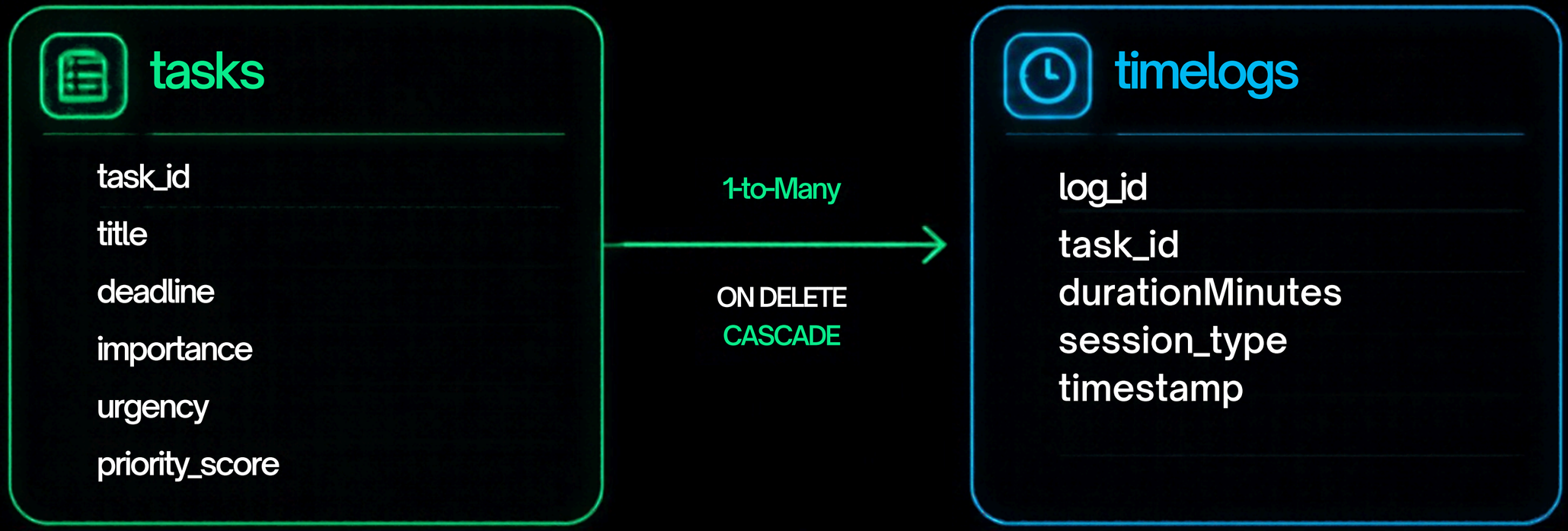
Simple on Top. Robust Underneath.

Separation of UI, Business Logic & Database



Built on Rock-Solid Relational Storage

MySQL 8.0 schema designed for fast queries & session tracking



SQL-Injection Proof
(PreparedStatement)

Auto-Cleanup
(Cascade Delete)

Sub-10ms Queries

Engineering with Pure Core Java

01
OOP Architecture

Clean abstraction with Prioritizable Interface

Inheritance via AbstractTask

Encapsulation throughout

Custom InvalidTaskException

Interfaces | Inheritance | Encapsulation

02
Smart Collections

Java PriorityQueue<Task>

Custom Comparator for O(log N) triage

Powerful Streams API

PriorityQueue | Comparator | Streams

03
Web & Concurrency

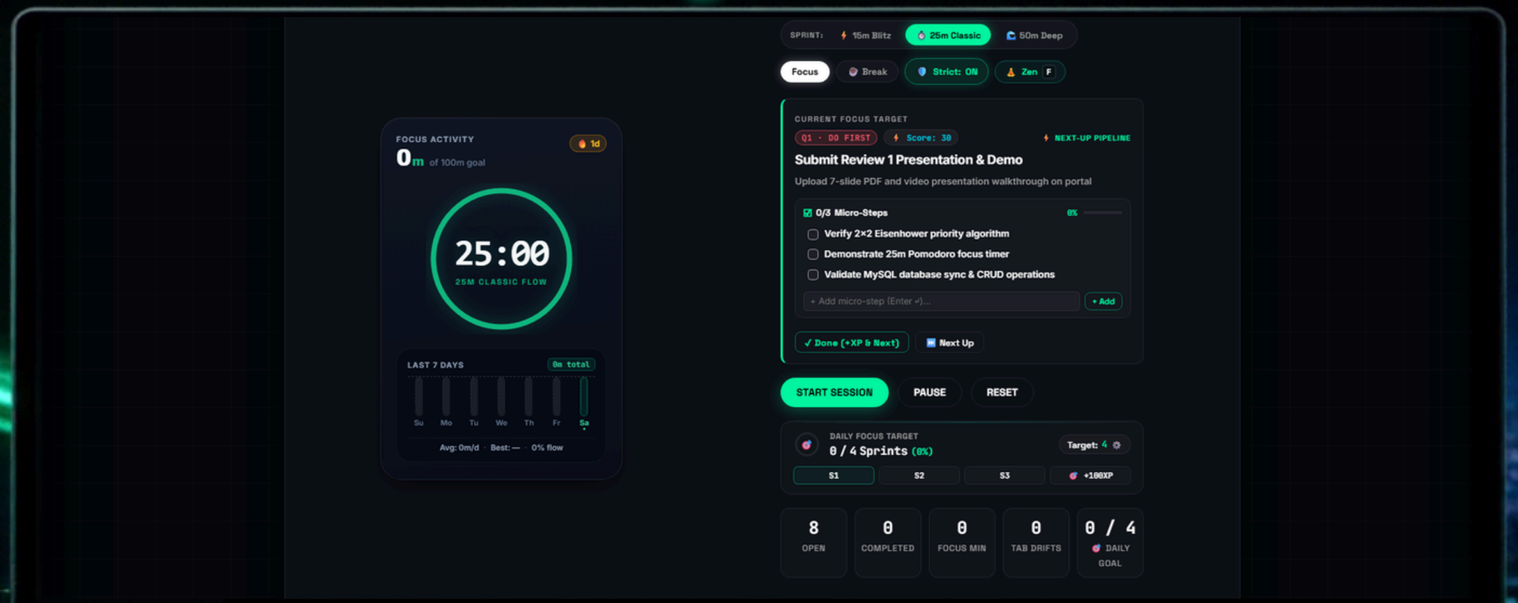
Native HTTP Exchange handlers (Servlets)

Background timer threads

Singleton JDBC connection

Servlets | Threads | JDBC

From Prototype to Daily Driver



What's Live Today

Full 2x2 Matrix
25:00 Audio Pomodoro
MySQL Sync
XP Leveling Engine

Next Milestones

Real-time multiplayer co-study rooms
Automated PDF summary exports
Cloud deployment